

ENDG 319

Probability, Statistics and Machine Learning

Lecture 5: Counting Methods

Fundamental principle • Permutations • Combinations

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From probability rules to counting outcomes

RECALL: FINITE, EQUALLY LIKELY OUTCOMES

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S} = \frac{|A|}{|S|}$$

In Lecture 4

List a small sample space and identify the event.

In Lecture 5

Count outcomes systematically when the list is large.

Counting gives the numbers; equal likelihood justifies the probability ratio.

Learning outcomes for Lecture 5

1

Count a sequence

Apply the fundamental counting principle to successive choices.

2

Count ordered arrangements

Use factorials and permutations when positions or roles matter.

3

Count unordered groups

Use combinations when only group membership matters.

4

Calculate a probability

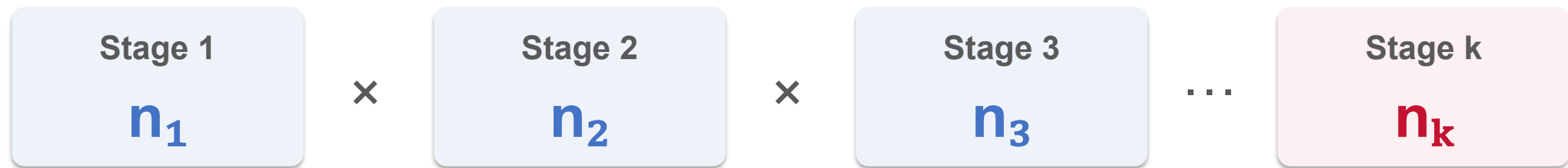
Count favourable and total outcomes using the same convention.

First ask: “What makes two outcomes different?”

The fundamental principle of counting

A task is completed through k successive operations.

For each valid sequence of earlier choices, stage i has n_i available choices.

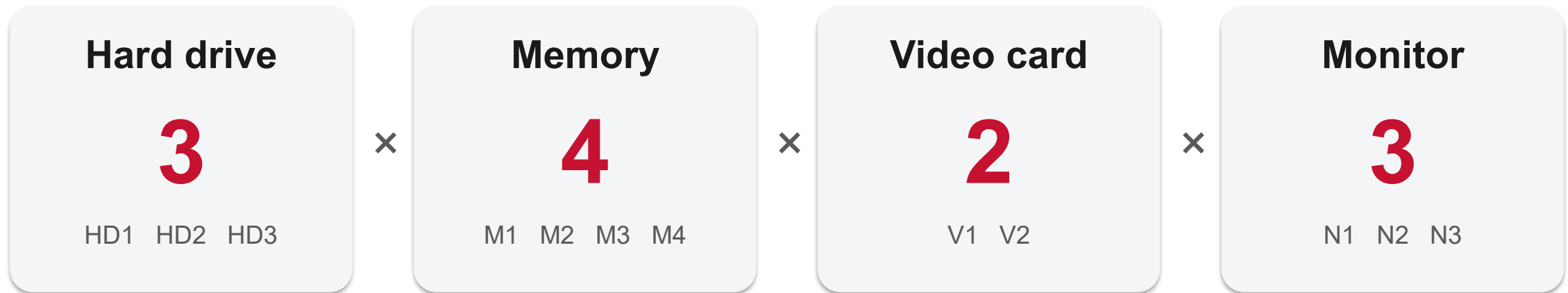


$$N = n_1 \times n_2 \times \dots \times n_k$$

Multiply the numbers of choices for the successive stages.

Example 4: How many computer configurations?

Choose one option from each component category.

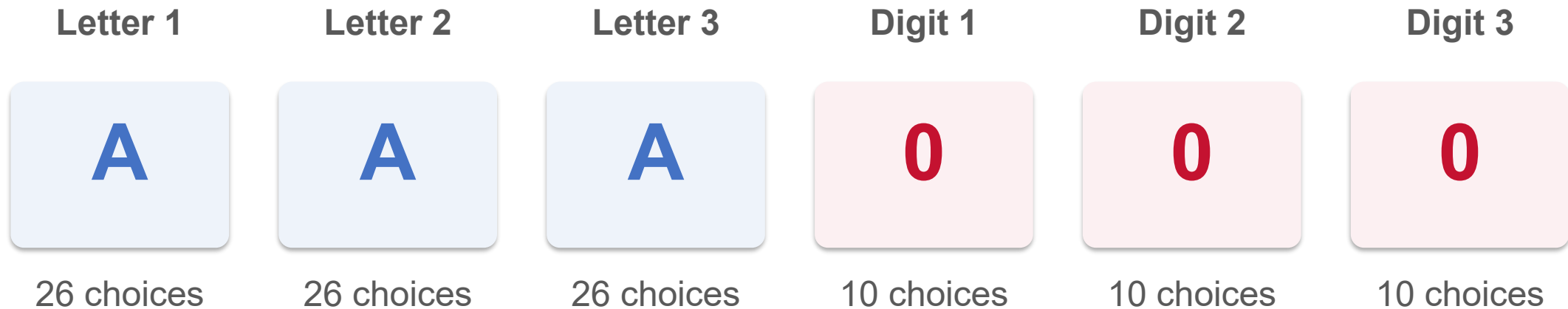


$$3 \times 4 \times 2 \times 3 = 72 \text{ configurations}$$

Assumption: every component option can be paired with every option at the other stages.

Example 5: Three letters followed by three digits

MODEL: A–Z AND 0–9; REPETITION AND ZEROS ARE ALLOWED

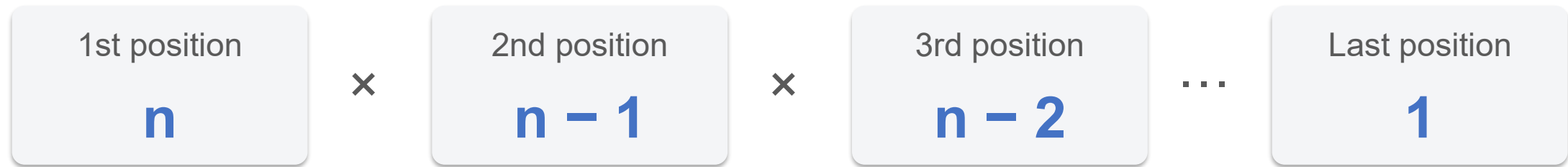


$$N = 26^3 \times 10^3 = 17,576,000$$

A plate is an ordered string. Reusing a character does not reduce later choices.

Permutations: Arrange all n distinct objects

A permutation is an ordering of a collection of objects.



$$N = n! = n(n - 1)(n - 2) \cdots 2 \cdot 1$$

$$4! = 24$$

$$5! = 120$$

$$0! = 1$$

All objects are distinct; each is used exactly once.

Permutations: Choose and order k of n objects

ORDER MATTERS • DISTINCT OBJECTS • NO REPETITION

k ordered positions:

$$\boxed{n} \times \boxed{n-1} \times \boxed{n-2} \cdots \boxed{n-k+1}$$

$${}_n P_k = \frac{n!}{(n-k)!} \quad 0 \leq k \leq n$$

The last factor is $n - k + 1$, not $n - k$.

A small check: Two letters selected from A, B, C

Selected group	Distinct ordered arrangements	
{A, B}	AB	BA
{A, C}	AC	CA
{B, C}	BC	CB

$${}_3P_2 = 3 \times 2 = 6$$

AB and BA are different permutations; neither repeats a letter.

Five runners: Award gold, silver, and bronze

Five runners enter a race. The top three receive different medals.



$${}_5P_3 = 5 \times 4 \times 3 = 60$$

The medal positions are different. Assume no ties and no repeated winner.

Example 6: Explain the counting method first

THINK • DISCUSS • JUSTIFY

A. Five people in a line

Five people stand in line at a movie theatre.

How many different orders are possible?

B. Three lifeguard stations

Five lifeguards are available for three distinct stations.

How many assignments are possible?

For each problem: identify n , identify k , and explain why order matters.

Example 6: Worked solutions

A. Arrange all five people

$$n = 5, k = 5$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

120 orders

B. Choose and assign three

$$n = 5, k = 3$$

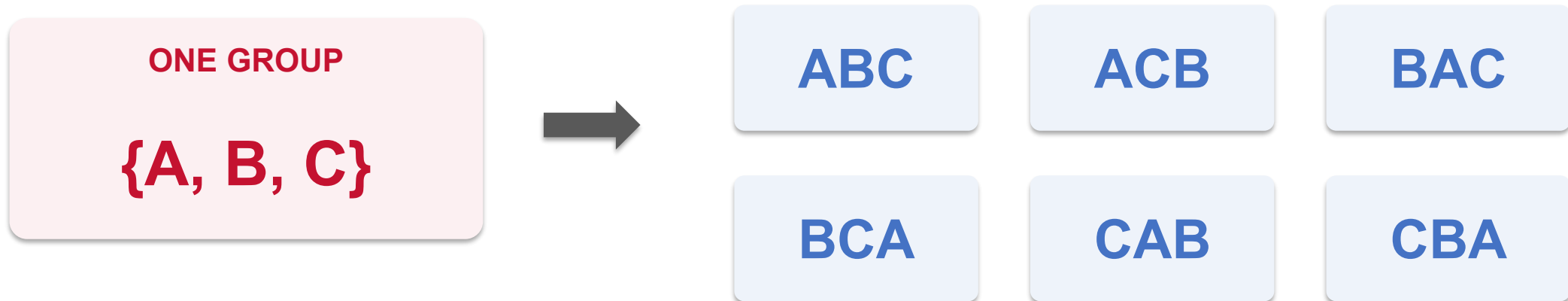
$${}_5P_3 = 5 \times 4 \times 3$$

60 assignments

Different roles or positions make the outcome ordered.

Combinations: Only the selected group matters

A combination is a selected group of objects, without regard to order.



$3! = 6$ orderings of the same group

Changing the order does not create a new combination.

Choosing three of A, B, C, D, E

TEN DISTINCT GROUPS • ORDER IGNORED

{A, B, C}

{A, B, D}

{A, B, E}

{A, C, D}

{A, C, E}

{A, D, E}

{B, C, D}

{B, C, E}

{B, D, E}

{C, D, E}

10 groups × 6 orderings per group = 60 permutations

To count the groups, divide 60 ordered arrangements by $3! = 6$.

Why the combination formula divides by k!

EVERY GROUP OF K DISTINCT OBJECTS HAS K! ORDERINGS

${}_nP_k$
ordered arrangements

÷

$k!$
orderings per group

$${}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Use combinations for unordered selections without repetition.

Same people, different questions

Choose a group of three

Only the identities of the selected people matter.

$${}_5C_3 = 10$$

Assign three different medals

The identity of the person in each medal position matters.

$${}_5P_3 = 60$$

10 groups × 3! arrangements per group = 60 ordered assignments

Example 7: Five identical door prizes

Thirty people attend an event. Five different people receive identical prizes.

ORDER DOES NOT MATTER

$${}_{30}C_5 = \frac{30!}{5!25!} = \frac{30 \times 29 \times 28 \times 27 \times 26}{5 \times 4 \times 3 \times 2 \times 1}$$

142,506 different groups

Identical prizes: count recipient groups, not the order of the announcement.

Choose a method by describing one outcome

One complete outcome describes...	Method	Count
A sequence of component or character choices	Fundamental principle	$n_1 \times n_2 \times \cdots \times n_k$
All n distinct objects in order	All-object permutation	$n!$
k of n distinct objects in order	Permutation	$n! / (n - k)!$
An unordered group of k distinct objects	Combination	$n! / [k!(n - k)!]$

For permutations and combinations here: no repeated object; $0 \leq k \leq n$.

Example 8: Select four bulbs for a lamp

A box contains 8 good bulbs and 2 burned-out bulbs.

SELECT FOUR DIFFERENT BULBS AT RANDOM; ORDER DOES NOT MATTER



$A = \{\text{all four selected bulbs are good}\}$

Model: each 4-bulb subset of the 10 individual bulbs is equally likely.

Example 8: Count the sample space and the event

All possible selections

Choose 4 from all 10 bulbs.

$${}_{10}C_4 = 210$$

Favourable selections

Choose 4 from the 8 good bulbs.

$${}_8C_4 = 70$$

$$P(A) = \frac{70}{210} = \frac{1}{3} \approx 0.3333 \approx 33.3\%$$

Use the same kind of outcome in the numerator and the denominator.

Four checks that prevent counting mistakes

1 Define one complete outcome

A computer order needs one option from every category.

2 Ask whether order or roles matter

Gold/silver/bronze and lifeguard stations are distinguishable.

3 Check whether an object can repeat

Plate characters can repeat; medal winners cannot.

4 Count consistently for probability

For the bulbs, count 4-item subsets in both numerator and denominator.

Explain the outcome definition before you write the counting expression.

Concept check: Find the counting error

EACH PROPOSED METHOD BELOW IS INCORRECT. EXPLAIN WHY.

A. Computer configurations

$$3 + 4 + 2 + 3$$

B. Medal assignments

$$5 \times 5 \times 5$$

C. Identical-prize groups

$$(30 \text{ choose } 5) \times 5!$$

D. All four bulbs good

$$P(A) = 8 / 10$$

Write the correct expression and explain what the incorrect one counts.

Concept check: Correct method and reasoning

A $3 \times 4 \times 2 \times 3 = 72$

Choose one option from each of four stages.

B $5 \times 4 \times 3 = 60$

A runner cannot fill more than one medal position.

C $30 \text{ choose } 5 = 142,506$

Identical prizes make announcement order irrelevant.

D $(8 \text{ choose } 4) / (10 \text{ choose } 4) = 1/3$

Count 4-bulb subsets, not individual good bulbs.

A correct answer includes both the expression and the reason it applies.

Lecture 5: Key takeaways

Successive choices

Count each stage.

$$n_1 \times n_2 \times \cdots \times n_k$$

Ordered arrangements

Distinct objects; no repetition.

$${}_nP_k = \frac{n!}{(n - k)!}$$

Unordered groups

Distinct objects; no repetition.

$${}_nC_k = \frac{n!}{k!(n - k)!}$$

Equally likely outcomes: $P(A) = \text{favourable count} / \text{total count}$.